The Router Is the Mechanism: Markup Floors, Steering, and the Design of AI Inference Marketplaces

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Abstract

AI inference marketplaces now route trillions of tokens per month across ~90 providers serving identical open-weight models, and the largest of them, OpenRouter, documents its allocation rule in one sentence: select a provider with probability proportional to $1/\text{price}^2$. A natural question to ask is: what market does this rule *buy*? We take the rule seriously as a mechanism and answer in three parts. (i) *The routing exponent is an inverse temperature, and the market sits near a phase transition.* Weights p^{-a} induce logit (Gibbs) demand; symmetric equilibrium is $p^* = c\,a(n-1)/(a(n-1)-n)$, which diverges along $a(n-1) = n$. The deployed exponent $a = 2$ places every two-provider market—the modal market in our five-minute panel of the largest live marketplace—in the condensed phase: equilibrium at the menu ceiling, and with the *measured* end-user elasticity (−0.05), a finite duopoly price of 41× marginal cost. An entry-proof markup floor $1/a$ survives any number of entrants: at $a = 2$, free entry stops at a 100% markup, forever. (ii) *The platform's own steering is a collusion device.* Our randomized probe panel measures the router penalizing a cheapest quote with a recent price cut (selection 3.9% vs. 23.3%, weight multiplier $\theta \approx 0.17$ for ~7 days). We prove this penalty deters undercutting for every agent with effective patience below $\delta^\dagger = 0.9895$—the measured θ sits five-fold inside the deterrence threshold—and taxes a perpetual micro-adjuster's flow by ~5×. In our calibrated, pre-registered-and-validated simulation, switching the measured penalty on flips a learning undercutter to the price ceiling in 4/4 markets and raises the flow-dominant bottom-of-book quote by up to 81%. (iii) *The mechanism is repairable, and the objectives disagree.* In a two-type world calibrated to observed capital structure (owned data-center vs. GPU-spot-dependent providers), welfare rises in the exponent while, counterintuitively, platform ad-valorem revenue *falls*—the router operator is paid to keep the market soft. We propose and evaluate, theoretically and with learning agents, three implementable repairs: a thickness-adaptive exponent $a^*(n) = n/(\ell^*(n-1))$ that pins the Lerner index at a target across market sizes; verified-quality weighting $w = q^b p^{-a}$ with q measured by our deployed evaluation probes, which undoes the quality-shading adverse selection that price-only weights create (closed-form threshold $b^* \approx 0.6$–2); and fee decoupling (per-request rather than ad valorem), which removes the platform's incentive to prefer high prices. All findings are for one marketplace (the largest), one buyer tier, and a registered re-estimation window; scope is stated throughout. Our results suggest that the observed pathologies of open inference markets—administered menus, focal-price atoms, futile undercutting—are equilibrium responses to routing-rule design choices, and that modest, implementable changes to the rule would repair them. The lever is the mechanism, not the conduct.

1 Introduction

Routing marketplaces. When an intermediary both *quotes* the market and *clears* it, the intermediary's algorithm is the demand curve. This is an old story in market microstructure—payment for order flow, last look in FX, the curvature of an automated market maker—and it is now the story of AI inference. On OpenRouter, the largest open inference marketplace, a buyer sends a request naming a model; a router selects which of n competing providers serves it. The default selection rule is documented in one sentence: filter recently failing providers, then choose with probability $\propto 1/\text{price}^2$. Nothing else a provider does moves demand except through that rule (and one measured exception we return to at length). So the game providers play is not Bertrand, not search, not bargaining: it is pricing against a softmax.[1]

What the data forces us to ask. We spent four months instrumenting this market at five-minute resolution (prices, congestion, realized flow, GPU spot books, randomized routing probes) and this paper asks the mechanism-design question the data forces: *is the observed price structure—rigid administered menus, a 45% atom of providers tied exactly at the model author's price, undercutters whose cuts appear futile, sustained 1.3–10× same-model dispersion—what the deployed rule should produce in equilibrium?* The answer is yes, in an uncomfortably strong sense: the deployed parameters sit in the region where theory predicts *exactly* the observed pathologies, and the platform's own quality-protective steering is, mathematically, the instrument that stabilizes them.

Temperature, not taste. The physics framing is not decoration. Weights p^{-a} are a Gibbs measure over providers with energy $a \log p$; the exponent is an inverse temperature. High temperature (small a) melts price competition—allocation ignores prices, so why cut them? Low temperature (large a) freezes allocation onto the cheapest provider—Bertrand, with its fragility. In between, there is a *critical line* $a(n-1) = n$ on which the symmetric equilibrium price diverges. The deployed $a = 2$ puts $n = 2$ markets exactly on the line. One does not operate a power plant at criticality by accident and call it a pricing strategy; but one might do it by choosing a "reasonable-looking" load-balancing rule without solving the game it induces. That, we will argue, is what happened—and because the

[1] The analogy to AMM curvature is exact in spirit: an AMM designer choosing invariant curvature trades price sensitivity against inventory risk [11]; a router choosing a trades price discipline against allocation concentration. Both are one-parameter families of demand curves imposed by an intermediary; both parameters are typically chosen by vibes.

rule is *documented*, every parameter in our theory is measured or published, not estimated from a structural model.[2]

Prior work. Calvano et al. [2] showed Q-learners sustain supra-competitive prices in a logit market they invented; we work in a logit market the platform *published*. Johnson–Rhodes–Wildenbeest [7] designed steering that breaks seller collusion but did not measure a deployed rule; Demirer et al. [8] measured this marketplace's dispersion and elasticity but did not model the router. None of these connects a documented allocation rule, a measured steering parameter, and calibrated counterfactuals; that connection is the paper.

Contributions. (1) A complete equilibrium analysis of inverse-power routing with captive and near-captive demand: interior formula, phase structure, an entry-proof Lerner floor, and the measured-elasticity duopoly price of $41c$ (§3). The static skeleton is Anderson–de Palma–Thisse logit pricing [1]; the content is the mapping—the "differentiation parameter" is a published platform constant, and its placement is on the critical line. (2) A theory of the measured steering penalty as a platform-imposed asymmetric menu cost: closed-form deviation target, deterrence threshold, patience boundary, perpetual-cutter tax (§4). To our knowledge the first steering-*supports*-collusion result with an empirically measured steering parameter (Johnson–Rhodes–Wildenbeest [7] study the inverse design). (3) A behaviorally calibrated, pre-registered, validated multi-agent environment in which every counterfactual in this paper is run (§5); its validation gate includes an untargeted moment—the simulated flow elasticity emerges from the router alone and matches the panel. (4) A design section with teeth (§6–7): two-type (data-center vs. spot) welfare/revenue/viability frontier over the exponent; a quality game showing price-only weights make shading dominant and verified-quality weights repair it at a small, computable b^*; a thickness-adaptive exponent; fee decoupling. Each proposal is evaluated with learning agents, not just first-order conditions.

2 The market and the data

OpenRouter routes buyer requests to ~90 providers serving shared open-weight models. Our capture (public repository `orcap`) records, at 5-minute grain: per-provider-per-model price menus; congestion telemetry (utilization, rate limits, latency, deranking); realized daily token flow per provider-model (demand shares); the vast.ai GPU spot order book (the input market for providers without owned capacity); provider capital structure (a confidence-flagged registry: hyperscaler DC / funded neocloud / own-silicon / small startup); and two active instruments: (a) a **randomized routing probe panel**—hourly one-token requests under default and pinned policies with randomized arm order, identifying the router's realized selection behavior for our buyer tier; (b) a **daily evaluation probe battery**—graded public benchmark items and deterministic prompts, per pinned provider, yielding verified per-provider quality signals (used in §7).

Stylized facts (established in the empirical companion under pre-registration discipline; all replicated in the repository): four split-sample-validated pricing species—*anchor adopters* (quote exactly the model author's price on ≥80% of days; 25% of pairs; 83–89% out-of-sample persistence), *static undercutters* (median −0.41 log, near-zero repricing), *active undercutters* (>1 change/day micro-adjusters, ~sub-1% flow), *premium* (+0.34 log, rigid); a tie atom at the author price 3.4× its grid-mechanical null, with author *identity* not special under a multiset-preserving null (focality without salience); routing-level flow elasticity −0.78 against end-user elasticity −0.05 (a 20× wedge); and the steering fact central to §4: among position-zero default-policy selections, a cheapest provider with a price cut in the trailing week is chosen 3.9% of the time versus 23.3% without ($\theta \approx 0.17$, memory ≈7 days).[3]

3 Equilibrium: temperature, criticality, and the markup floor

n providers post prices $p_i \in (0, \bar{p}]$ ($\bar{p}$ the menu ceiling); marginal costs c_i; a unit mass of requests arrives per epoch; the router selects i first with probability $s_i = p_i^{-a}/\sum_j p_j^{-a}$. Profits $\pi_i = s_i(p_i - c_i)$. With logits $-a \log p_i$ this is softmax: *the router is a logit demand system with inverse temperature a.*

Lemma 3.1. *$\partial s_i/\partial p_i = -(a/p_i)\, s_i(1 - s_i)$: the own-price share elasticity is $-a(1 - s_i)$—bounded by a no matter how many rivals.*

Lemma 3.2 (Single crossing). *π_i is strictly quasiconcave on $(c_i, \bar{p}]$; the best response is the unique root of $h(p) \coloneqq a(1-s_i(p))\frac{p-c_i}{p} = 1$, or the ceiling if $h(\bar{p}) \le 1$.*

Theorem 3.3 (Phase structure and floor). *With symmetric costs c: (i) if $a(n-1) > n$ (and $\bar{p} > p^*$), the unique symmetric equilibrium is*

$$p^* = c\,\frac{a(n-1)}{a(n-1)-n}; \tag{1}$$

(ii) if $a(n-1) \le n$, it is the menu ceiling $\bar{p}$—at $a = 2$, precisely $n \le 2$; (iii) for $a > 1$, ANY interior equilibrium—asymmetric costs, any n, any outside option—satisfies $(p_i - c_i)/p_i > 1/a$, i.e., $p_i > c_i\, a/(a-1)$, and the floor is tight as $n \to \infty$.

Proof. Appendix A; every closed form here and below is enforced in continuous-integration tests against continuum numerics (13 tests). □

Corollary 3.4 (Measured elasticity). *With aggregate demand $D \propto P^{\varepsilon}$ (inclusive-value index P, ε the end-user elasticity), the symmetric FOC is $p/(p-c) = a(n-1)/n + |\varepsilon|/n$. At the measured $\varepsilon = -0.05$ and deployed $a = 2$: duopoly $p^* = 41c$. Captivity is the $\varepsilon \to 0$ limit, not an assumption we need.*

In other words, the router's smoothing is itself the market power: no conduct is required for a 100% markup to persist. Three readings. *The floor is entry-proof:* Lemma 3.1 bounds share elasticity by a, so no amount of entry raises the elasticity a provider faces above a; at $a = 2$, markups never fall below 100%. Softmax smoothing—adopted for uptime insurance and exploration—*is* product differentiation,

[2] Structural IO estimates a logit differentiation parameter from data and argues identification. Here the platform *publishes* it. This is the rare setting where the mechanism-design counterfactual requires no demand estimation at all: the demand curve is in the documentation.

[3] The probe protocol is deliberately minimal: hourly one-output-token requests (`max_tokens: 1`, temperature 0) under randomized arm order, with the served provider read from the marketplace's generation-metadata endpoint. The entire panel costs a few dollars a month; the identification comes from the randomization, not the spend.

manufactured by the platform. *The critical line is populated:* the modal market in our panel has 1–3 providers; $a = 2$ puts $n = 2$ on the line, and the observed menus-at-ceiling behavior (premium species, adopters at the author's list price) is the condensed phase.[4] *The wedge is structural:* Corollary 3.4 turns our measured 20× elasticity wedge into the theorem's parameter—near-captive end-user demand is exactly why realistic menus bind before equilibrium does.

4 Steering: the measured penalty is a deterrence device

Model the measured behavior as: a provider whose price is below its price at any of the last M epochs has weight $\theta\, w(p)$. (At symmetric profiles, "any recent cutter" and the measured "cheapest with a recent cut" coincide—any strict cut makes you cheapest; our calibrated experiments run both variants.)

Theorem 4.1 (Deterrence). *At a symmetric profile q with rival weight $W = (n-1)q^{-2}$: (i) the flagged deviant's optimal cut is $p_{\text{dev}} = c + \sqrt{c^2 + \theta/W}$; (ii) undercutting is unprofitable for a one-epoch optimizer iff $\theta \le \theta^*$, the unique root of $v(\theta^*) = (q-c)/n$; across calibrated configurations $\theta^* \in [0.81, 1.0]$—the measured $\theta = 0.17$ sits deep inside; (iii) a one-time cut held forever is deterred iff $\delta \le \delta^\dagger = 0.9895$ (calibrated parameters; $\delta^M \le 0.93$); (iv) an agent cutting at least once per M epochs is permanently flagged and pays share tax $\theta/(\theta + n - 1)$ vs. $1/n$— 4.9× at measured values.*

This bound demonstrates that the measured penalty is roughly five times stronger than the weakest penalty that would already deter undercutting. The economic reading: the penalty—presumably deployed as bait-and-switch protection—is an *asymmetric menu cost on price cuts only, imposed by the platform.* It converts the routing game into one where holding high prices is robustly optimal for any realistically patient agent, and it specifically taxes the high-frequency undercutting technology we observe dying in the data (the registered cut-frequency decay watch). Johnson–Rhodes–Wildenbeest [7] designed steering to *break* seller collusion; the deployed steering is their inverse, with a measured parameter.[5]

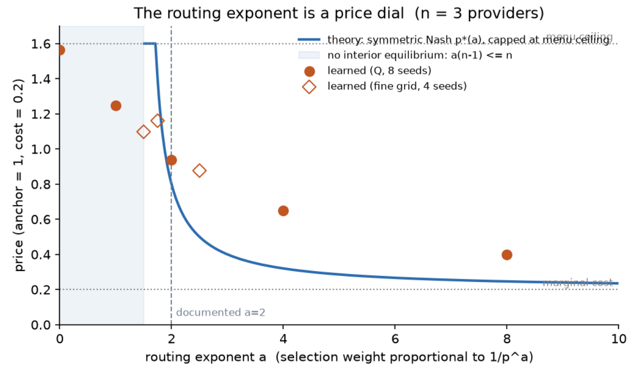

Figure 1: The routing exponent is a price dial ($n = 3$). Theory Nash curve (capped at the menu ceiling; shaded region: no interior equilibrium, $a(n-1) \le n$) with converged Q-learning prices overlaid. The deployed $a = 2$ sits at the edge of the collapse; learning friction smooths the critical singularity.

Table 1: Two-type frontier across the routing exponent. Flow-price ∝ ad-valorem platform revenue; welfare = $v - \sum_i s_i c_i$.

a	p_L	p_S	flow-price	welfare	spot profit
1.5	0.99	1.60	1.25	1.731	0.155
2	0.51	1.10	0.65	1.802	0.049
3	0.27	0.76	0.30	1.875	0.005
4	0.20	0.67	0.20	1.895	0.001
6	0.15	0.60	0.15	1.900	~0

5 The calibrated environment and its validation gate

Counterfactuals about mechanisms require agents, and assuming Bertrand agents would beg every question. Our environment fits the observed species (margins, hazards, rationing slopes from the panel's train window; costs as identified bands from the capital-tier registry and the GPU book with its walk-the-book impact curve) and *must pass a pre-registered validation gate before any counterfactual runs*: ten moments computed by identical code on simulated and observed panels, distance ≤ 0.04, with the moments explicitly split into calibration echoes and genuinely emergent tests. The confirmatory run (20 seeds, deterministic replication) passes at distance 0.019; the two emergent moments carry the weight—*flow elasticity* −0.65 ± 0.35 *vs.* −0.78 *observed with no fitted allocation parameter* (the documented rule alone generates the measured demand response), and adopter-atom out-of-sample persistence 0.83 vs. 0.834. Learning counterfactuals then show (all pre-specified, gated, seeded): the exponent is a price dial (Figure 1; uniform 1.53 ⟶ a=2: 0.94 ⟶ floor at $a \ge 8$, with learning friction regularizing the critical singularity); learners never rediscover micro-adjustment and form anchor ties endogenously; and the steering counterfactual of §4—penalty on flips the learning undercutter to the ceiling in 4/4 calibrated markets (broad rule) and raises the flow-dominant bottom-of-book quote +81% where the strict measured conditional binds.

6 Heterogeneous capital and the objective frontier

Providers are not symmetric. The registry splits them into owned-capacity types (hyperscaler DCs, own silicon; low marginal cost, high fixed cost—the "reserved instance" world) and spot-dependent types (rent H100s on a thin order book; marginal cost includes walking the book, +18–52% at realistic fill sizes). Take $c_L = 0.10$, $c_S = 0.50$ (the calibrated band endpoints), $n_L = 2$, $n_S = 3$, and solve the asymmetric FOC system $p_i/(p_i - c_i) = a(1 - s_i)$ across a (Table 1).

[4] The analogy is quantitative, not decorative: the fitted species heterogeneity plays the role of quenched disorder—frozen provider types over which the annealed price dynamics run—and the critical line is where the quenched free energy stops being self-averaging across markets.

[5] A last-look analogy is exact: in FX, dealers who get cut off after adverse quote updates learn to quote wide and stable. Here the router plays the liquidity consumer who punishes aggressive quoting.

The objectives disagree, and the platform is conflicted. Welfare (allocative: flow sorted to low-cost capacity) is increasing in a; ad-valorem platform revenue is *decreasing* in a—a platform charging a percentage fee is paid to keep the market soft. And spot-dependent viability collapses in a: the sharp-routing world is a world where only data-center owners survive, which prices resilience at zero until the first correlated outage.[6]

Robustness (cost bands). The welfare-increasing-in-a ordering holds at all four corners of the identified cost bands ($c_L \in \{0.05, 0.25\}$, $c_S \in \{0.3, 0.7\}$): monotone in every case. *Resilience-adjusted welfare.* With probability ρ per epoch all owned capacity fails simultaneously; if the spot fleet has exited (profit below fixed cost $F_S = 0.01$), those requests are lost at value v. At $\rho = 2\%$: adjusted welfare is 1.802 (a=2, spot viable) vs. 1.855–1.860 ($a \geq 4$, spot exited)—sharp routing still wins, but the gap narrows fourfold; the ranking flips at $\rho \gtrsim 5\%$. The honest design statement: softness is welfare-justified as outage insurance only for correlated-outage intensities above ~5% per epoch; below that, the right architecture is sharp routing *plus* priced capacity commitments (§7), not a soft exponent that pays every provider an insurance premium whether or not insurance is delivered.

E-MECH1 (the frontier with learners). Replacing the FOC with Q-learning agents (5 seeds/arm) confirms the ordering—learned welfare 1.66 (uniform) → 1.797 (a=2) → 1.856 (adaptive $a^*(n)$=6.25) → 1.898 (WTA); learned flow-weighted price 1.58 → 0.68 → 0.43 → 0.40; spot-type profit 0.64 → 0.157 → 0.011 → −0.000 (winner-take-all drives spot types below cost)—with one addition theory alone would not have exhibited so starkly: the deployed pair ($a = 2$ + the measured cut-penalty) is *simultaneously the worst arm on welfare* (1.66, tied with uniform routing) *and the best for ad-valorem platform revenue* (flow price 1.60—the menu ceiling; all learners at the cap). The steering does not merely blunt the exponent's discipline; it deletes it. The platform's deployed configuration is the revenue-maximizing corner of the frontier we traced.

7 Mechanism repairs

Thickness-adaptive temperature. Theorem 3.3 inverts: to hold the Lerner index at target ℓ^* in an n-provider market, set $a^*(n) = n/(\ell^*(n-1))$. At $\ell^* = 0.2$: $a^* = 10$ (n=2), 7.5 (n=3), 6.25 (n=5), → 5.56. One line of router code; kills the critical line (no market sits in the condensed phase); keeps softmax's exploration/insurance properties (unlike winner-take-all). Cost: §6's viability column—sharp temperatures starve spot-dependent capacity, so ℓ^* should embed the resilience value, or be paired with capacity commitments below.

Verified-quality weighting. Price-only weights create adverse selection in *quality*: serving a quantized/degraded variant saves cost Δ at buyer-invisible quality damage d, and under $w = p^{-a}$ shading is strictly dominant (share unchanged, cost falls). Weight $w = q^b p^{-a}$ instead, with q verified: shading multiplies weight by $(1-d)^b$, and the symmetric all-high profile is an equilibrium iff $b \geq b^*$ solving

$$\frac{1}{n}(p-c) \;=\; \frac{(1-d)^b}{(1-d)^b + n - 1}\,(p - c + \Delta). \tag{2}$$

At calibrated values (n=3, d=0.2, Δ=0.08): $b^* = 0.63$—a *modest* quality exponent suffices, and b^* falls with n. The required q is not hypothetical: our deployed evaluation probes already produce per-provider graded-accuracy and output-consistency scores daily. E-MECH2 (5 seeds, 600k epochs): learned high-quality share rises 0.27 (b=0, the deployed quality-blind rule) → 0.40 (b=0.5) → 0.67 (b=1) → 0.53 (b=2). The direction matches the theory—quality weighting shifts learned play toward quality provision, with the largest jump crossing b^*—but the bifurcation is noisy rather than sharp: the quality margin ($\Delta = 0.08$) is small relative to price-learning noise in the doubled action space, and at $b = 2$ quality-weight and price competition interact. We report this as directional confirmation of the closed-form threshold, not a sharp learnability boundary.

Fee decoupling. Ad-valorem fees make the platform's objective ∝ flow-weighted price (§6). A per-request (or per-token-served) flat fee makes platform revenue proportional to *volume*, aligning the platform with allocative efficiency and removing its stake in the price level. This is the one-line governance repair that makes the adaptive exponent incentive-compatible *for the platform.*

Steering redesign and commitment contracts. Replace the asymmetric cut-penalty with (i) symmetric change-penalties (a true menu cost—removes the directional deterrence of Theorem 4.1 while keeping stability protection), or (ii) quality/uptime-conditioned steering only (penalize serving failures, not price cuts—the stated goal without the collusive side effect). For spot-dependent viability under sharp temperatures: capacity-commitment contracts (the reserved-instance analog): providers pre-commit admitted capacity at posted prices and are penalized for reneging—converting resilience from an implicit subsidy paid via soft routing into an explicit priced product.

8 Related work

Anderson–de Palma–Thisse [1] (logit oligopoly); Calvano et al. [2], Klein [3], Asker et al. [4], Abada–Lambin [5], Hansen–Misra–Pai [6] (algorithmic pricing/collusion— our demand system nests Calvano's, so their apparatus transfers); Johnson–Rhodes–Wildenbeest [7] (platform steering against collusion; we document and analyze its deployed inverse); Demirer et al. [8] (OpenRouter empirics: dispersion, elasticity); PriLLM [9] (Stackelberg routing theory, no calibration or steering); Fish et al. [10] (LLM pricing agents; an LLM-agent tier exists in our codebase; running it at confirmatory scale is future work and not claimed here); Angeris–Chitra [11] (curvature as an intermediary's demand-shape choice); token-pricing screening [12]; metering and covert quantization audits (the practices §7 prices). None combine a documented allocation rule, measured steering, calibrated behavioral heterogeneity, and mechanism repairs evaluated with learning agents.

9 Limitations

Theory: symmetric-equilibrium characterization (the floor is the asymmetric statement); captive baseline with measured-ε corollary;

[6]Practitioner reading: this is the routing-market version of the PFOF debate. The intermediary's fee base is the price level; sharpening execution quality cuts its own revenue. The fix (§7) is the same as in brokerage: decouple the fee from the price.

capacity non-binding (rationing lives in the empirical companion). Measurement: θ from one conditional slice of one buyer tier; panel short (30-day re-estimation registered, auto-reopening); costs are identified sets (all quantitative claims reported across bands). Simulation: tabular Q at Calvano hyperparameters; expected-allocation training (exact under non-binding capacity); LLM-agent tier built, not confirmatory. The quality game's d and Δ are calibrated to our eval-probe detection scale, not to a demand-side damage estimate.

10 Practitioner takeaways

For router operators: (1) publish n-adaptive temperatures, not one exponent—$a^*(n) = n/(\ell^*(n-1))$ is one line; (2) never penalize price cuts asymmetrically—you are operating a cartel enforcement device; penalize failures, not prices; (3) weight by verified quality with even a small b—your own eval probes suffice; (4) charge flat fees if you want to be believed about (1)–(3). For providers: under the deployed rule, undercutting is (measurably) taxed and micro-adjustment is dominated—the observed adopter/premium behavior is the rational response, which is precisely the problem. For regulators: conduct-based tools will find nothing here; every elevated price in this paper is a static Nash equilibrium of a published rule.

11 Conclusion

The router is the demand curve, and at the deployed parameters it is a demand curve that cannot discipline anyone: duopolies sit at the menu ceiling ($41c$ at the measured outside elasticity), entry stops at a 100% markup, and the platform's own anti-cut steering pays providers to hold prices high. Every one of these statements follows from a published constant and a measured parameter, which is precisely why they are fixable: an n-adaptive temperature, verified-quality weights, and flat fees are each one line of code or one line of a fee schedule. The honest caveat is scale—one marketplace, one buyer tier, a registered re-estimation window that can reopen every calibrated claim—but the mechanism analysis needs none of the usual structural leaps, at least at first glance: the demand curve is in the documentation.

A Proofs

Lemma 3.1. $s_i = w_i/(w_i + W_{-i})$, $w_i = p_i^{-a}$; $dw_i/dp_i = -(a/p_i)w_i$, so $ds_i/dp_i = -(a/p_i)s_i(1 - s_i)$. □

Lemma 3.2. $d\pi_i/dp_i = s_i\,[1 - h(p_i)]$ with h a product of positive strictly increasing functions and $h(c_i) = 0$; single crossing gives strict quasiconcavity and best-response uniqueness. □

Theorem 3.3. (i) Impose $s = 1/n$ in $h(p) = 1$; uniqueness from monotonicity of h at the symmetric profile; the corner $\bar{p}$ fails when $h(\bar{p}) > 1$ (profitable downward deviation). (ii) $h(q) < a(n-1)/n \le 1$ at any symmetric q, so raising is strictly profitable below the ceiling; quasiconcavity rules out global downward deviations at $\bar{p}$. (iii) $1 - s_i < 1$ in the FOC $\Rightarrow$ Lerner $> 1/a$; tightness as $s_i \to 0$. □

Corollary 3.4. With $D = D_0 P^\varepsilon$ and $\partial \log P/\partial \log p_i = s_i$: $d\log \pi_i/d\log p_i = \varepsilon s_i - a(1 - s_i) + p_i/(p_i - c)$; set to zero at $s = 1/n$. Same single-crossing argument. (The inclusive-value aggregator is a modeling choice; any aggregator with $\partial \log P/\partial \log p_i = s_i + o(s_i)$ gives the same FOC to first order.) □

Theorem 4.1. (i) The FOC of $\theta p^{-2}(p - c)/(\theta p^{-2} + W)$ reduces to $Wp^2 - 2Wcp - \theta = 0$. (ii) $v(\theta)$ is continuous, strictly increasing (envelope), $v(0) = 0$; intermediate value. (iii) direct present-value computation; the root $\delta^\dagger$ is numerically certified. (iv) substitute equal prices with one weight scaled by θ. □

All closed forms are enforced by 13 continuous-integration tests against continuum numerics.

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